IRREDUCIBLE-SURJECTION CONDITIONS FOR UNIQUE ORDER-PRESERVING ISOMETRY EXTENSIONS

AGENT LANE 16

Status

This is a candidate full literal answer to the final Problem in Tanmoy Paul and T. S. S. R. K. Rao, *Order-preserving unique Hahn-Banach extensions*, arXiv:2504.03386 [1]. The problem asks for conditions under which an order-preserving isometric embedding

$$\Phi : A(K) \longrightarrow C(K)$$

has a unique extension to an into order-preserving isometry

$$T : C(K) \longrightarrow C(K).$$

The theorem below gives such conditions in the unital setting used throughout the source paper: the state map associated with Φ is required to be an irreducible continuous surjection. This is not a classification of every possible embedding, but it is a direct set of conditions answering the source request as stated.

Source problem

The source problem appears on PDF page 14, in the final displayed Problem of [1]. A crop of the statement is included below.

> **Problem 4.8.** *Operator theoretic versions of the results considered here are still open. It is a folklore result that for an infinite dimensional Choquet simplex K, whose extreme boundary is not closed, the identity map $i : A(K) \to C(K)$ has no norm preserving extension to $C(K)$. To see this, note that existence of an extension implies that $A(K)$ is the range of a contractive projection on $C(K)$. It follows from Theorem 5 in Section 10 of* [9] *(see also the classification scheme in* [11]*), since $A(K)$ is the range of a contractive projection in a space of continuous functions, $\partial_e A(K)_1^* \cup \{0\}$ is a weak*-compact set. Since $\mathbf{1} \in A(K)$, 0 can not be an accumulation point. Now it is easy to see that $\partial_e K$ is a closed set, giving the required contradiction. In view of the operator versions from* [15]*, it would be interesting to find conditions under which an order-preserving isometric embedding $\Phi : A(K) \to C(K)$ has unique extension to an into order-preserving isometry from $C(K)$ to $C(K)$?*

The final sentence asks:

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> In view of the operator versions from Rao, it would be interesting to find conditions under which an order-preserving isometric embedding $\Phi : A(K) \to C(K)$ has unique extension to an into order-preserving isometry from $C(K)$ to $C(K)$.

Definitions

Let K be a compact convex subset of a locally convex real topological vector space. Let $A(K)$ be the real Banach space of continuous affine real-valued functions on K, and let $C(K)$ be the real Banach lattice of continuous real-valued functions on K, both with the supremum norm.

Definition 1. A continuous surjection $\alpha : K \to K$ is called irreducible if no proper closed subset $L \subsetneq K$ satisfies $\alpha(L) = K$.

Every homeomorphism is irreducible, and the identity map is irreducible. There are also non-injective irreducible surjections in general topology, so the condition is broader than invertibility.

Idea of the Proof

The source problem is an into-isometry problem for $C(K)$-spaces, so the right structural input is Holsztynski's into form of the Banach-Stone theorem. It says that any linear isometry $T : C(K) \to C(K)$ is forced to be a composition operator on some closed subset $L \subset K$ that still sees the whole domain space: there is a continuous surjection $\beta : L \to K$ such that $Tf(t) = f(\beta(t))$ for $t \in L$. If T is positive and unital, no sign factor appears.

For an order-preserving unital embedding $\Phi : A(K) \to C(K)$, evaluation at each $t \in K$ gives a state of $A(K)$, hence a point $\alpha(t) \in K$; so $\Phi(a)(t) = a(\alpha(t))$. If T extends Φ, then on the Holsztynski set L we have

$$a(\beta(t)) = Ta(t) = \Phi(a)(t) = a(\alpha(t))$$

for every affine function a. Since affine functions separate points, $\beta(t) = \alpha(t)$ on L. But $\beta(L) = K$, so $\alpha(L) = K$; irreducibility forces $L = K$. Thus T is the composition operator $f \mapsto f \circ \alpha$ everywhere.

The theorem

Lemma 1 (State-map form of unital positive embeddings)**.** *Let $\Phi : A(K) \to C(K)$ be a unital positive linear map. Then there is a unique continuous map $\alpha_\Phi : K \to K$ such that*

$$\Phi(a)(t) = a(\alpha_\Phi(t)) \qquad (a \in A(K),\ t \in K).$$

If Φ is an isometry, then the closed convex hull of $\alpha_\Phi(K)$ is K. If α_Φ is onto, then Φ is an isometry.

Proof. For each $t \in K$, the functional $a \mapsto \Phi(a)(t)$ is positive and takes the constant 1 to 1. It is therefore a state of the order-unit space $A(K)$. As recalled in the source paper's preliminaries, the state space of $A(K)$ is affinely homeomorphic to K, so there is a unique $\alpha_\Phi(t) \in K$ such that $\Phi(a)(t) = a(\alpha_\Phi(t))$ for all $a \in A(K)$. Continuity of α_Φ follows because the functions in $A(K)$ separate points and each $t \mapsto \Phi(a)(t)$ is continuous.

For $a \in A(K)$,

$$\|\Phi(a)\| = \sup_{t\in K} |a(\alpha_\Phi(t))| = \sup_{x\in\alpha_\Phi(K)} |a(x)|.$$

Thus Φ is isometric exactly when $\alpha_\Phi(K)$ is norming for $A(K)$. By the Hahn-Banach separation theorem for compact convex sets, this is equivalent to the closed convex hull of $\alpha_\Phi(K)$ being K. In particular, if α_Φ is onto then Φ is an isometry. □

Theorem 1. *Let $\Phi : A(K) \to C(K)$ be a unital order-preserving linear map, and let $\alpha = \alpha_\Phi$ be the associated state map. Assume that $\alpha : K \to K$ is an irreducible continuous surjection. Then Φ is an order-preserving isometric embedding, and it has a unique positive linear isometric extension $T : C(K) \to C(K)$. This extension is*

$$T(f) = f \circ \alpha \qquad (f \in C(K)).$$

Proof. Since α is onto, the lemma gives that Φ is an isometry. The operator

$$C_\alpha : C(K) \to C(K), \qquad C_\alpha(f) = f \circ \alpha,$$

is positive, unital, and isometric, and it extends Φ. Thus existence is immediate.

Let $T : C(K) \to C(K)$ be any positive linear isometry extending Φ. Since $1 \in A(K)$ and $\Phi(1) = 1$, we have $T1 = 1$. By Holsztynski's theorem for into isometries of $C(K)$-spaces [2], there are a closed subset $L \subset K$ and a continuous surjection $\beta : L \to K$ such that

$$Tf(t) = f(\beta(t)) \qquad (f \in C(K),\ t \in L).$$

The usual unimodular sign factor in the real Banach-Stone representation is 1 here because $T1 = 1$.

For $a \in A(K)$ and $t \in L$,

$$a(\beta(t)) = Ta(t) = \Phi(a)(t) = a(\alpha(t)).$$

Continuous affine functions separate points of K, hence $\beta(t) = \alpha(t)$ for every $t \in L$. Since $\beta(L) = K$, it follows that $\alpha(L) = K$. Irreducibility of α forces $L = K$. Therefore, for every $f \in C(K)$ and $t \in K$,

$$Tf(t) = f(\beta(t)) = f(\alpha(t)).$$

So $T = C_\alpha$, proving uniqueness. □

Corollary 1 (Canonical inclusion)**.** *For every compact convex K, the canonical unital inclusion*

$$A(K) \subseteq C(K)$$

has exactly one positive linear isometric extension $C(K) \to C(K)$, namely the identity operator.

Proof. Here $\alpha = \mathrm{id}_K$, which is an irreducible continuous surjection. The theorem applies. $\square$

Corollary 2 (Homeomorphic composition embeddings)**.** *For every compact convex K and every homeomorphism $h : K \to K$, the embedding $\Phi(a) = a \circ h$ from $A(K)$ into $C(K)$ has the unique positive linear isometric extension $T(f) = f \circ h$. No metrizability or point-peak hypothesis is needed.*

Verification and novelty notes

The proof is analytic and uses no numerical computation. The decisive external ingredient is Holsztynski's into-isometry theorem for spaces of continuous functions; the rest is the standard state-space representation of $A(K)$, which is also recalled in the source paper.

Before this upgrade, the run indexes were searched for arXiv:2504.03386, the source title, and core phrases around order-preserving isometric embeddings, unique positive isometry extensions, irreducible surjections, Banach-Stone, and Holsztynski. The only exact target hit was the prior partial packet `2504.03386_peakpoint_unique_positive_isometry_extension`. A bounded web search for exact phrases from the source problem and close variants did not find a direct later answer. The theorem should therefore be reviewed as a new literal-answer packet, with the caveat that it gives conditions rather than a full classification.

Human-review recommendation

Review as a candidate full literal answer to the source problem. The main points to check are: (i) the source context justifies the unital hypothesis, (ii) Holsztynski's theorem is being applied in the correct real $C(K)$ into isometry form, and (iii) the phrase "find conditions" is accepted as answered by the irreducible-surjection condition rather than requiring a classification of all possible embeddings.

References

[1] Tanmoy Paul and T. S. S. R. K. Rao, *Order-preserving unique Hahn-Banach extensions*, arXiv:2504.03386, 2025. `https://arxiv.org/abs/2504.03386`

[2] W. Holsztynski, *Continuous mappings induced by isometries of spaces of continuous functions*, Studia Mathematica 26 (1966), 133–136.